# SteamLens: A Cross-Lingual Analysis System for Aspect-Level Review Disparities and Controversy Evolution on Steam

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**Track B: Demonstration**

## Abstract

Simplified Chinese and English reviews of the same Steam game often report markedly different recommendation rates, yet aggregate rates cannot say what players object to, when a gap widens, or whether similar complaints cluster in time. We propose SteamLens, a cross-lingual review analysis system that turns recommendation-rate gaps into inspectable and traceable textual evidence by combining aspect-based sentiment analysis, near-duplicate expression detection, and temporal alignment with verified update events.

## 1 Project Title

SteamLens: A Cross-Lingual Analysis System for Aspect-Level Review Disparities and Controversy Evolution on Steam.

## 2 Team Members & Responsibilities

This is a single-author project. I am responsible for the problem definition, API and literature research, compliant data acquisition and cleaning, the label specification, automatic pre-annotation as weak supervision (Bu et al., 2021) and its manual verification, construction of the independent human test set, implementation of all core NLP modules, experimental evaluation, the interactive demo, the final report, and the classroom presentation. All open-source tools, pretrained models, and AI assistance will be documented with their sources and specific uses.

## 3 Background & Motivation

Steam titles whose Simplified Chinese recommendation rate falls clearly below other language groups are called "ring-region positive review" cases. *Slay the Spire 2* and *Final Fantasy VII Rebirth* are the two case studies: preliminary small-sample requests to the official review API return summaries consistent with a Chinese–English gap, and because such summaries shift with time and filters, the analysis uses snapshots recording collection time and the full query condition. Differing player priorities and in-game experiences, differing standards for recommending, and concentrated expression after updates and community diffusion all offer plausible explanations that aggregate rates cannot separate, and they identify neither what players object to nor when a divergence appears, both of which are targets of aspect-based sentiment analysis (Pontiki et al., 2016).

SteamLens therefore measures discussion content, its change over time, and the concentration of similar wording. Its units are the *Simplified Chinese review group* and the *English review group*, a linguistic grouping that does not identify nationality, region, or native language. A Steam recommendation vote is a binary "recommend to others" choice that may reflect gameplay, price, update policy, and community experience at once, not a uniform-scale quality score (Valve Corporation, 2024). The system targets players, developers, and course audiences who want to compare feedback across language groups.

## 4 Problem Definition / Research Questions

A user selects a Steam game and specifies a time range and a sampling cap. The system acquires Chinese and English reviews and their metadata through the official API, including game identity, language, text, recommendation vote, creation and update timestamps, playtime at review time, and collection time, and it also reads previously saved fixed snapshots. Models trained during development then produce aspect and sentiment predictions, and the system reports comparable Chinese–English statistics, aspect and sentiment distributions, temporal trends, and clusters of similar expressions with representative evidence as interac-

tive charts. Event-window analysis is a case-level extension that takes event dates verified against official announcements. The analysis unit is the review, and a single review may involve several aspects, with its recommendation vote and per-aspect sentiments stored separately.

Our research questions are as follows. **RQ1 (aspect focus)**: within the same game and time window, which aspects do the two groups discuss, and how do aspect mention rates and negative-attitude rates differ, where mention rates use all qualifying reviews as the denominator and per-aspect negative rates use only reviews mentioning that aspect. **RQ2 (controversy evolution)**: around a major update, how do review volume, recommendation rate, and the distribution of negative topics shift in each group, and does the gap widen or narrow with the event. **RQ3 (expression convergence)**: in windows where negative reviews concentrate, are near-duplicate phrasings, similar-expression clusters, and explicit community references also more concentrated, separating "discussing the same issue" from "using near-identical wording" so that shared game terminology alone does not indicate imitation.

Our working hypothesis is that cross-lingual recommendation-rate differences co-occur with differences in discussed aspects, reactions to events, and expression concentration. The project addresses these questions with aspect-based sentiment analysis, near-duplicate detection, and temporal event alignment, implemented on a frozen multilingual encoder (Devlin et al., 2019; Conneau et al., 2020) with aspect and sentiment classification heads alongside rule and TF-IDF baselines, following recent cross-lingual aspect-based sentiment work (Šmíd et al., 2025). The scope covers text-based evidence for these differences; nationality inference, personal profiling, cross-platform identity linkage, and causal claims about social impact lie outside the project.

## 5 Expected Outcomes

The project will deliver a runnable Chinese–English review analysis prototype that lets a user select a game and, within a bounded window and sampling cap, complete review collection, aspect sentiment prediction, near-duplicate detection, temporal comparison, and interactive visualization. It will include a classroom-ready fixed-snapshot demo, a capped online entry point, and a demonstration backup keeping live and cached data distinct, plus reproducible code, run instructions and dependency records, sampling records, the label specification, and independent human test results. Model quality will be reported as micro-F1 and Macro-F1 for aspect detection and as Accuracy, Macro-F1, and per-class Precision, Recall, and F1 for aspect sentiment, separately for Chinese and English so that cross-lingual model differences are not read as differences in player priorities.

The analysis is expected to produce verifiable observations in the two main cases, such as which demands increased after a given update and where the two groups diverge most, together with a small-scale generalization check on a game that did not participate in training or tuning.

## 6 Project Timeline

The proposal deadline is 12 October 2026; the midterm, final, and presentation dates remain subject to course announcement, so the schedule uses relative weeks. Week 1 validates pagination, time windows, and online sampling limits, builds data snapshots and sampling rules for the main cases, completes roughly 50–100 trial annotations, and assesses pre-annotation quality and verification cost. Week 2 fixes the label specification and data split, isolates the independent test data, prepares verified training and validation labels alongside pseudo-labels, and runs the rule and TF-IDF baselines. Week 3 completes the first version of the aspect sentiment and near-duplicate modules and connects fixed-snapshot analysis to a simple interface. Week 4 performs model selection and limited ablations on the validation set, finishes the human test labels and the new-game selection criteria, freezes the configuration, and runs the human test evaluation and the cross-lingual error analysis. Week 5 completes the generalization check, system acceptance for both modes, and the case and sensitivity analyses, and Week 6 prepares the report, reproduction instructions, contribution materials, and the presentation.

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